higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 4

1 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 True
1.2 False
1.3 True
1.4 True
1.5 False

(5 × 1) **[5]**

QUESTION 2

2.1 C
2.2 E
2.3 F
2.4 H
2.5 B

(5 × 1) **[5]**

QUESTION 3

3.1
- Water stops flowing in plastic waste, making it the ideal breeding place for mosquitos.
- Plastic is landfilled, which costs money and takes up a lot of space.
- Toxic and environmentally harmful chemicals are sometimes used or produced during plastic production and these can cause environmental or human health problems, damage to the soil, pollute the air including surface water sources and can increase global warming.
- Plastics that are discarded into the environment can be ingested by animals resulting in death, or become entrapped in the plastic waste preventing them from escaping, flying, swimming or eating.
- Synthetic polymers are non-biodegradable thus can accumulate in the environment.
- Polymers/Plastics are made from petrochemical monomers which are produced from non-renewable natural sources. (Any 4 × 1) (4)

3.2 3.2.1
- Tough
- Resistance to chemicals
- Resistance to water absorption (Any 2 × 1) (2)

3.2.2 LDPE is manufactured by a high-pressure process at (1 000 to 3 000 atm)

HDPE is manufactured by a low-pressure process at (1 to 200 atm) (4)

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3.3 3.3.1
- Water (demineralised)
- VCM (vinyl chloride monomer)
- Initiator (diacetyl peroxide)
- Suspending agent (PVA) (Any 3 × 1) (3)

3.3.2
- Production of the vinyl chloride monomer (VCM) from raw materials
- Polymerisation of VCM to PVC (2)

3.4
- The raw materials, solvent, monomers and initiator are fed to the reactor.✓
- The conversion is 100%; there is no need for the monomers to be recovered.✓
- After polymerisation is complete, the rubber solution is stored in a tank.✓
- From the tank, the cement is coagulated.✓ After coagulation, the rubber is stripped with steam and hot water to remove the solvent which creates a rubber slurry.✓
- The slurry is dewatered before a belt dryer is used to remove the residual water.✓ (6)

3.5
- Footwear (shoe soles and heels)
- Conveyor belts and rubber drive belts
- Car tyres and tyre retreads (as an SBR coating)
- Coated paper (E-SBR)
- Building applications such as a sealing and binding agent as an alternative to PVA paint (Any 3 × 1) (3)

3.6
- Fillers and fibres
- Pigments and colourants
- Catalysts
- Lubricants
- Plasticisers
- Binders (Any 3 × 1) (4)

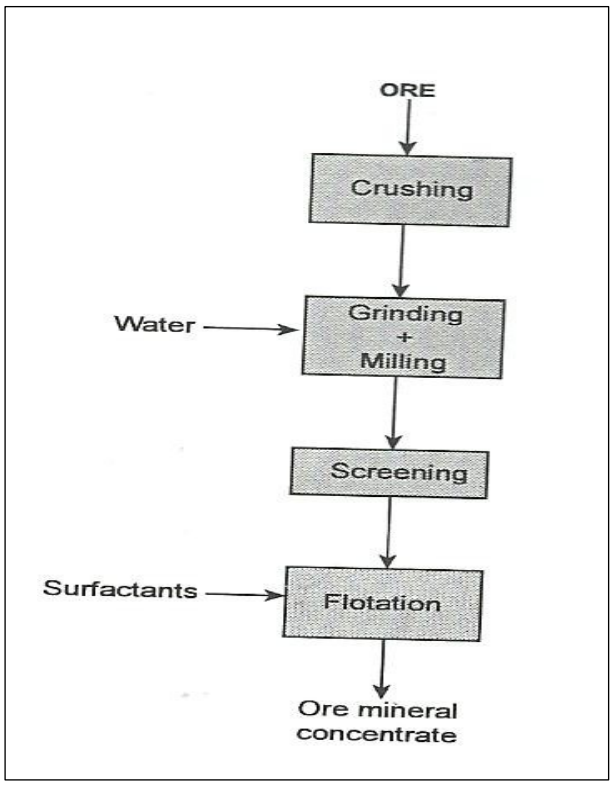
[28]

QUESTION 4

4.1
- Amplats
- Implats
- ARM platinum
- Phalaborwa Mining Company Limited (4)

4.2 Hydrometallurgy/use of solution (aqueous) chemistry in the recovery of metals (1)

4.3



(8)

4.4 4.4.1 Gold (Au)

4.4.2 Palladium (Pd)

4.4.3 Platinum (Pt)

(3 × 1) (3)

4.5 A Crushing and milling
B Roasting in a kiln
C Precipitation
D Filtration
E Electric arc furnace
F Ammonium sulphate
G Ferrovanadium (7 × 1) (7)

[23]

QUESTION 5

5.1 Sasol (1)

5.2 5.2.1
- Production of synthetic fertilizers
- Production of explosives
- Production of cold packs

5.2.2
- Production of sulphur
- Production of sulphuric acid and hydrochloric acid
- Passivation of coil metallurgy

(2 × 3) (6)

5.3
- Petrol
- Diesel
- Paraffin/Jet fuel
- Liquid petroleum gas (LPG)
- Fuel oil
- Asphalt

(Any 4 × 1) (4)

5.4
- Shell SA
- BP SA

(2)

5.5 Sour oil contains a higher percentage of hydrogen sulphide than sweet oil. (2)

5.6 5.6.1 This is the process where cracking takes place in the presence of hydrogen.

5.6.2 This process uses heat to break down the long-chained hydrocarbon into shorter ones.

5.6.3 This is the process of cracking hydrocarbons in the presence of a catalyst.

(3 × 2) (6)

5.7 It produced a large amount of solid coke, which was not needed. (1)

5.8
- Lower temperature and pressures are needed.
- It has a higher yield of petrol and other distillates.
- It has lower gas yields (low-value products).
- Higher yields of aromatics.
- Produces petrol with a higher octane number.

(Any 4 × 1) (4)
[26]

QUESTION 6

6.1 6.1.1
- Enzymes are reusable biological catalysts which assist in the completion of biochemical reactions✓ and at the end of the reaction they remain unchanged✓ (Any 2 × 1)

6.1.2
- Hormones are chemical messengers which initiate biological reactions ✓resulting in their chemical composition changing thus cannot be reutilized✓ (Any 2 × 1)

(2 × 2) (4)

6.2
- Adcock Ingram LTD
- Aspen Pharmacare
- Cipla Medpro (Any 2 × 1) (2)

6.3 A lauter tun, mash tun or mash filter can be used for lautering.✓ The mixture is transferred to this filter where the liquor, called wort, is allowed to pass through✓ and the grist solids are trapped.✓ Hot water is sprinkled into the mash to wash all the dissolved sugar out of the mash.✓ (4)

6.4 Cooling reduces enzyme inactivation✓ and chemical degradation during extraction.✓ (2)

6.5 It is a liquid containing nutrients for culturing a variety of micro-organisms.

or

A liquid nutrient medium which can support the growth of numerous microorganisms that are not very nutritionally demanding. (1)

[13]

TOTAL: 100

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